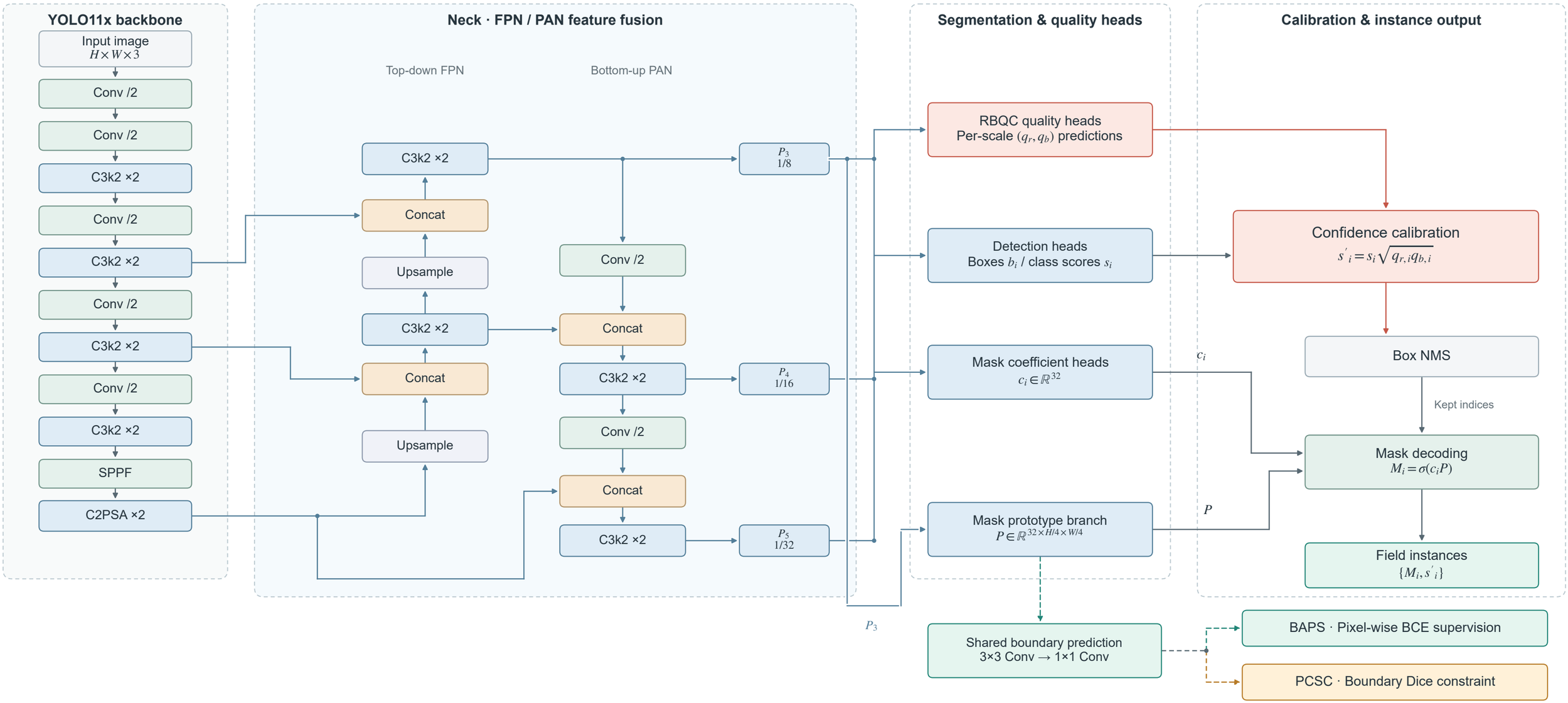
# FieldBCQ

Field instance segmentation with three improvement mechanisms

Training & inference

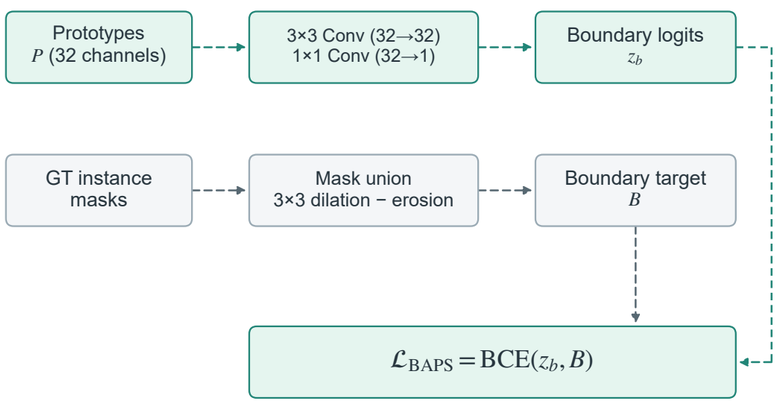
Training only

## (a) Overall network architecture



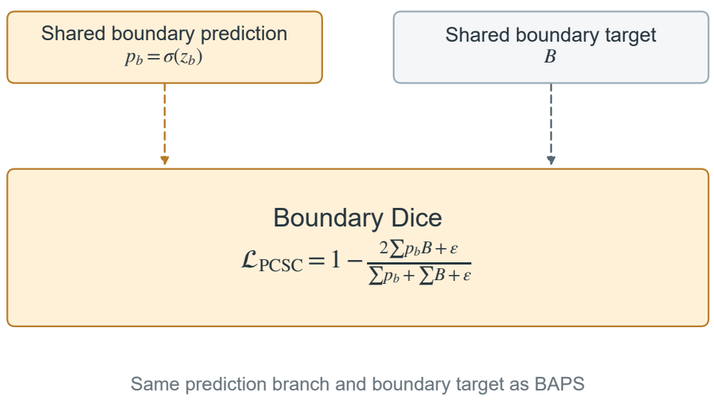

## (b) BAPS

Boundary-aware pixel supervision




## (c) PCSC

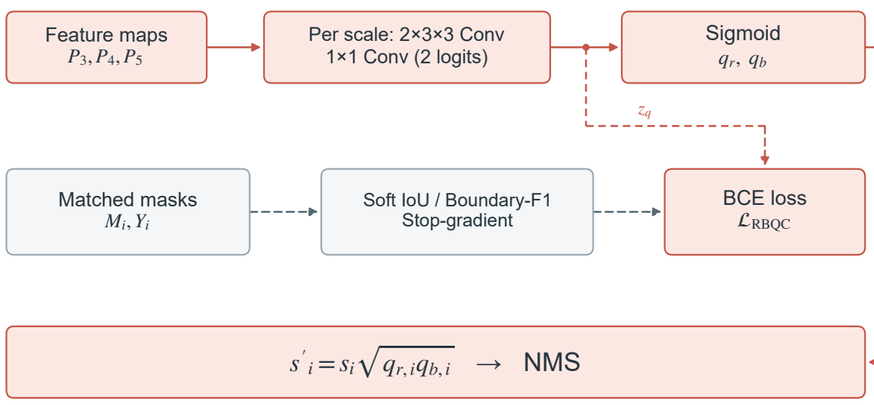
Parcel contour structural consistency




Same prediction branch and boundary target as BAPS

## (d) RBQC

Region-boundary quality calibration




$$\mathcal{L} = \mathcal{L}_{\text{YOLO-seg}} + \lambda_b(\mathcal{L}_{\text{BAPS}} + \mathcal{L}_{\text{PCSC}}) + \lambda_q \mathcal{L}_{\text{RBQC}}$$